DESCRIPTION OF COURSEWORK

Course Code	EEE314
Course Name	Digital System Design
Lecturer	Ir. Dr. Chua Ming Yam
Academic Session	2023/09
Assessment Title	Assignment: Task 2

A. Introduction/ Situation/ Background Information

FPGA stands for field-programmable gate array. It is an integrated circuit that implements code in hardware to execute a thousand times faster than in a processor. These circuits, or arrays, consist of configurable logic blocks (CLBs), memory, or other elements. FPGAs boast rapid growth over the past decade due to the wide range of applications they can handle. For example, it has become a major subject of research and utilization in many engineering fields like hardware acceleration, wireless body sensor networks, diagnostic imaging, electromedical segments, cardiac management, genomics/life science, computer architectures, and more. This course demonstrates how digital design and rapid prototyping could be facilitated by FPGAs and hardware description languages (Verilog)

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 1 Apply the knowledge and skills of logic designs into hardware description language

CLO 2 Design solutions for complex engineering problems in digital design and FPGA

CLO 3 Build digital systems using a modern EDA tool

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:

i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).

ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.

iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

1. This is an individual task. You are required to complete it on your own.
2. You are required to use Quartus Prime to complete this task.
3. Use the cover page provided for your report.
4. You are required to submit your report and your Quartus archived project file through Moodle.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Individual report	50%
2.	Quartus project	50%
	TOTAL	**100%**

F. Task(s)

This is an individual task that contributes to part of your coursework mark.

A 14-bit binary to BCD decoder module is given below.

```
module bin2bcdx4 (
      input [13:0] binary,
      output reg [3:0] x1000, x100, x10, x1
);

      integer i;
      always @(binary)
      begin
            // set x1000, x100, x10, and x1 to 0
            x1000 = 4'b0000;
            x100 = 4'b0000;
            x10 = 4'b0000;
            x1 = 4'b0000;

            for (i=13; i>=0; i=i-1)
            begin
                  // add 3 to columns >= 5
                  if (x1000 >= 5)
                        x1000 = x1000 + 3;
                  if (x100 >= 5)
                        x100 = x100 + 3;
                  if (x10 >= 5)
                        x10 = x10 + 3;
                  if (x1 >= 5)
                        x1 = x1 + 3;

                  // shift left
                  x1000 = x1000 << 1;
                  x1000[0] = x100[3];
                  x100 = x100 << 1;
                  x100[0] = x10[3];
                  x10 = x10 << 1;
                  x10[0] = x1[3];
                  x1 = x1 << 1;
                  x1[0] = binary[i];
            end
      end
endmodule
```

14-bit binary to BCD decoder

Using the start-up code below, create a top-level design named “bin2sevensegmentx4” that instantiates a “bin2bcdx4” module and four “BCD2SevenSegment” modules (from Task 1). The output of the “bin2bcdx4” module (x1000, x100, x10, and x1) shall be connected to the respective input of the “BCD2SevenSegment” module. A sample of the RTL diagram is shown in Figure 1.

```
module bin2sevensegmentx4 (
      input [13:0] binary,
      output [6:0] HEX0, HEX1, HEX2, HEX3
);

      wire [3:0] bcdx1000, bcdx100, bcdx10, bcdx1;

      // Your codes here

endmodule
```

Start-up code for “bin2sevensegmentx4”

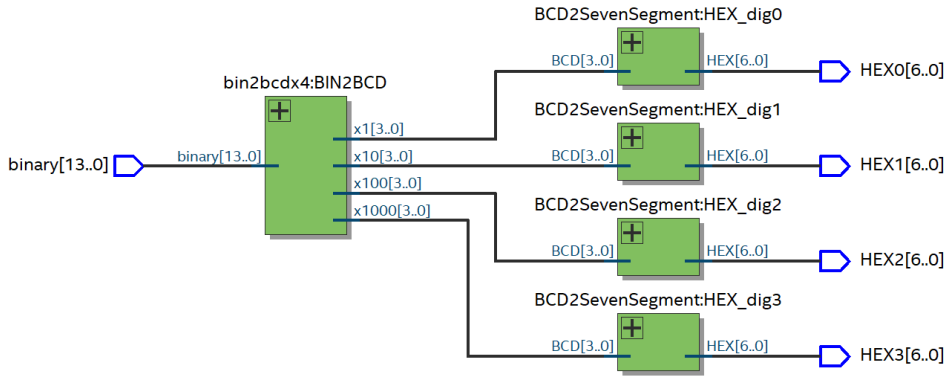


Figure 1: RTL diagram of top-level design for “bin2sevensegmentx4”

Verify your design using the DE10-Standard board.

Write a short report discussing the outcomes of your project. Submit your report and Quartus archived project file to the Moodle.

APPENDIX 1

MARKING RUBRICS

Component Title	Individual report				**Percentage (%)**		50	
Criteria	**Score and Descriptors**						**Weight (%)**	**Marks**
	Excellent (5)	**Good (4)**	**Average (3)**	**Fair (2)**	**Poor (1)**	**Very Bad (0)**		
Report contents (CLO2-PLO3-C4)	Well-written report	Requires minor improvement	Satisfactory contents	Requires major revision	Poorly written report, insufficient contents	Did not submit	40	
Language and formatting (CLO2-PLO3-C4)	Good language convention with neat formatting		Satisfactory language convention and formatting		Poor language / formatting	Did not submit	10	
						TOTAL	**50**	

Component Title	Quartus project				**Percentage (%)**		50	
Criteria	**Score and Descriptors**						**Weight (%)**	**Marks**
	Excellent (5)	**Good (4)**	**Average (3)**	**Fair (2)**	**Poor (1)**	**Very Bad (0)**		
Quartus project (CLO3-PLO5-P4)	Well working project with no issue	Complete project but with 1 or 2 issue(s)	Complete project but with > 2 issues	Incomplete project, majority parts were missing	Blank project / almost all parts were missing	Did not submit	50	
						TOTAL	**50**	